

Group Assignment 2

Group 22

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June 13, 2026

1. Consider a supervised learning problem where we assume that $Y|X$ is Poisson distributed. That is, the conditional density of $Y|X$ is given by

$$f_{Y|X}(y, x) = \frac{\lambda^y e^{-\lambda}}{y!}, \lambda(x) = \exp(\alpha x + \beta)$$

Here α is a vector (slope) and β is a number (intercept).

Follow the calculations from Section 4.2.1 to derive a loss that needs to be minimized with respect to α and β . Note: do we really need the factorial term?

Solution 1:

For n i.i.d. data points (X_i, Y_i) , the likelihood is:

$$L(\alpha, \beta) = \prod_{i=1}^n \frac{\lambda(X_i)^{Y_i} e^{-\lambda(X_i)}}{Y_i!}.$$

Taking the natural logarithm:

$$\ln L(\alpha, \beta) = \sum_{i=1}^n [Y_i \ln \lambda(X_i) - \lambda(X_i) - \ln(Y_i!)].$$

Negating gives the negative log-likelihood:

$$-\ln L(\alpha, \beta) = \sum_{i=1}^n [-Y_i \ln \lambda(X_i) + \lambda(X_i) + \ln(Y_i!)].$$

The term $\ln(Y_i!)$ does not depend on α or β , so it can be omitted for minimization. Thus, the loss function to minimize is:

$$\ell(\alpha, \beta) = \sum_{i=1}^n [\lambda(X_i) - Y_i \ln \lambda(X_i)].$$

Substituting $\lambda(X_i) = \exp(\alpha \cdot X_i + \beta)$:

$$\ell(\alpha, \beta) = \sum_{i=1}^n [e^{\alpha \cdot X_i + \beta} - Y_i(\alpha \cdot X_i + \beta)].$$

This is the loss function that must be minimized with respect to α and β .

2. Let $X_1, \dots, X_n$ be IID from $\text{Uniform}(0, \theta)$. Let $\hat{\theta} = \max(X_1, \dots, X_n)$. First, find the distribution function of $\hat{\theta}$. Then compute the bias($\hat{\theta}$), $SE(\hat{\theta})$, and $MSE_n(\hat{\theta})$.

Solution 2:

1. Distribution Function

For $0 < t < \theta$,

$$F_{\hat{\theta}}(t) = P(\hat{\theta} \leq t) = P(X_1 \leq t, \dots, X_n \leq t) = [P(X_i \leq t)]^n = \left(\frac{t}{\theta}\right)^n.$$

Hence the probability density function is

$$f_{\hat{\theta}}(t) = \frac{d}{dt}F_{\hat{\theta}}(t) = n \frac{t^{n-1}}{\theta^n}, \quad 0 < t < \theta.$$

2. Expected Value and Bias

The expected value of $\hat{\theta}$ is

$$E[\hat{\theta}] = \int_0^\theta t f_{\hat{\theta}}(t) dt = \int_0^\theta t \cdot n \frac{t^{n-1}}{\theta^n} dt = \frac{n}{\theta^n} \cdot \frac{\theta^{n+1}}{n+1} = \frac{n}{n+1} \theta.$$

Therefore, the bias is

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta = -\frac{\theta}{n+1}.$$

The estimator is negatively biased, it tends to underestimate θ .

3. Variance and Standard Error

Compute

$$E[\hat{\theta}^2] = \int_0^\theta t^2 f_{\hat{\theta}}(t) dt = \frac{n}{\theta^n} \int_0^\theta t^{n+1} dt = \frac{n}{n+2} \theta^2.$$

Thus,

$$\text{Var}(\hat{\theta}) = E[\hat{\theta}^2] - (E[\hat{\theta}])^2 = \frac{n\theta^2}{(n+1)^2(n+2)}.$$

The standard error is

$$SE(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})} = \frac{\theta\sqrt{n}}{(n+1)\sqrt{n+2}}.$$

4. Mean Squared Error

By definition,

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2.$$

Substituting the expressions above,

$$MSE(\hat{\theta}) = \frac{n\theta^2}{(n+1)^2(n+2)} + \frac{\theta^2}{(n+1)^2} = \frac{2\theta^2}{(n+1)(n+2)}.$$

Summary

$$\begin{aligned}
 F_{\hat{\theta}}(t) &= \left(\frac{t}{\theta}\right)^n, & 0 < t < \theta, \\
 E[\hat{\theta}] &= \frac{n}{n+1}\theta, \\
 \text{Bias}(\hat{\theta}) &= -\frac{\theta}{n+1}, \\
 \text{Var}(\hat{\theta}) &= \frac{n\theta^2}{(n+1)^2(n+2)}, \\
 SE(\hat{\theta}) &= \frac{\theta\sqrt{n}}{(n+1)\sqrt{n+2}}, \\
 MSE(\hat{\theta}) &= \frac{2\theta^2}{(n+1)(n+2)}.
 \end{aligned}$$

3. Consider the continuous distribution with density

$$p(x) = \frac{1}{2} \cos(x), -\frac{\pi}{2} < x < \frac{\pi}{2}$$

- (a) Find the distribution function F
- (b) Find the inverse distribution function F^{-1}
- (c) To sample using an Accept-Reject sampler, Algorithm 1, we need to find a density g such that $p(x) \leq Mg(x)$ for some $M > 0$. Find such a density g and find the value of M .

Solution 3:

(a) Distribution function F .

For $-\frac{\pi}{2} < x < \frac{\pi}{2}$,

$$F(x) = \int_{-\pi/2}^x \frac{1}{2} \cos t \, dt = \frac{1}{2} \left(\sin x - \sin\left(-\frac{\pi}{2}\right) \right) = \frac{1}{2}(\sin x + 1).$$

Hence,

$$F(x) = \begin{cases} 0, & x \leq -\frac{\pi}{2}, \\ \frac{1 + \sin x}{2}, & -\frac{\pi}{2} < x < \frac{\pi}{2}, \\ 1, & x \geq \frac{\pi}{2}. \end{cases}$$

(b) Inverse distribution function F^{-1} .

For $u \in (0, 1)$, set

$$u = \frac{1 + \sin x}{2} \implies \sin x = 2u - 1.$$

Therefore, the inverse CDF is

$$F^{-1}(u) = \arcsin(2u - 1),$$

with $F^{-1}(0) = -\frac{\pi}{2}$ and $F^{-1}(1) = \frac{\pi}{2}$.

(c) Choice of g and M for Accept-Reject sampling.

A convenient choice is the uniform density on the support:

$$g(x) = \frac{1}{\pi}, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}.$$

Compute

$$\frac{p(x)}{g(x)} = \frac{\frac{1}{2} \cos x}{1/\pi} = \frac{\pi}{2} \cos x,$$

whose maximum occurs at $x = 0$ and equals $\frac{\pi}{2}$.

Hence, we can take

$$M = \frac{\pi}{2},$$

so that $p(x) \leq Mg(x)$ for all x in the support.

4. Let $Y_1, Y_2, \dots, Y_n$ be a sequence of IID discrete random variables, where $P(Y_i = 0) = 0.1, P(Y_i = 1) = 0.3, P(Y_i = 2) = 0.2$, and $P(Y_i = 3) = 0.4$. Let $X_n = \max(Y_1, \dots, Y_n)$. Let $X_0 = 0$ and verify that $X_0, X_1, \dots, X_n$ is a Markov chain. Find the transition matrix P .

Solution 4:

Since $X_{n+1} = \max(X_n, Y_{n+1})$ and Y_i is an IID sequence, we have

$$P(X_{n+1}|X_n, X_{n-1}, \dots, X_0) = P(X_{n+1}|X_n)$$

Therefore, the Markov property holds and so it is a Markov chain.

Let the state space be $\{0, 1, 2, 3\}$ and let $p(x) = P(Y = x)$. We can see that

- If it stays at state i , the probability is $\sum_{x=0}^i p(x)$
- If it moves to state j where $j > i$, then the probability is $p(j)$
- If it moves to state j where $j < i$, then the probability is 0

Then the transitional matrix is

$$\begin{pmatrix} 0.1 & 0.3 & 0.2 & 0.4 \\ 0 & 0.4 & 0.2 & 0.4 \\ 0 & 0 & 0.6 & 0.4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

5. Let $X_1, \dots, X_n$ be IID from some distribution F that is unknown. Let $\hat{F}_n$ be the empirical distribution function, use this to find an estimate of the p quantile of F (call it q). Use Theorem 5.28 to find a confidence interval for q .

Solution 5

Let $X_1, \dots, X_n \stackrel{iid}{\sim} F$, and let $\hat{F}_n(x)$ be the empirical CDF:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq x\}}.$$

The p -quantile q of F is defined as

$$q = F^{-1}(p) = \inf\{x : F(x) \geq p\}.$$

A natural estimate of q is the empirical p -quantile:

$$\hat{q}_n = \hat{F}_n^{-1}(p) = \inf\{x : \hat{F}_n(x) \geq p\}.$$

In practice, if we sort the data $X_{(1)} \leq \dots \leq X_{(n)}$, we have

$$\hat{q}_n = X_{(\lceil np \rceil)}.$$

The Dvoretzky-Kiefer-Wolfowitz (DKW) inequality states:

$$\Pr \left(\sup_x |\hat{F}_n(x) - F(x)| > \epsilon \right) \leq 2e^{-2n\epsilon^2}.$$

For a confidence level $1 - \alpha$, we solve for ϵ :

$$2e^{-2n\epsilon^2} = \alpha \quad \implies \quad \epsilon = \sqrt{\frac{1}{2n} \ln \frac{2}{\alpha}}.$$

This gives a confidence interval for the p -quantile q :

$$\left[\hat{F}_n^{-1}(p - \epsilon), \hat{F}_n^{-1}(p + \epsilon) \right],$$

or equivalently using the order statistics:

$$\left[X_{(\lceil n(p-\epsilon) \rceil)}, X_{(\lceil n(p+\epsilon) \rceil)} \right].$$